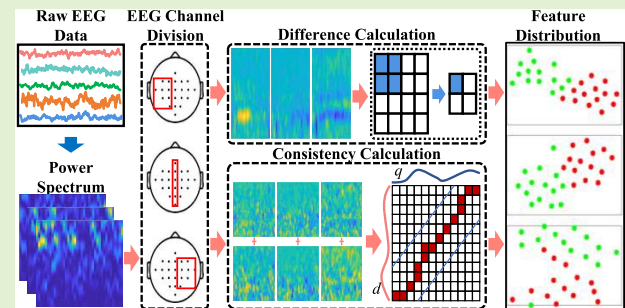


A Power Spectrum Pattern Difference-Based Time-Frequency Sub-Band Selection Method for MI-EEG Classification

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Abstract—Motor imagery-based brain-computer interface (MI-BCI) is considered to be the most promising technology, which can help patients with muscle disorders to carry out rehabilitation training and assist in activities of daily living. However, the time window and frequency band of the ERD/ERS pattern activated by motor imagery vary from person to person. In this study, we propose a power spectrum pattern difference-based time-frequency sub-band selection (PSPD) method, which can further improve the classification performance of MI EEG. To better analyze the sensorimotor rhythm, we divide the EEG channels and extract the average and maximum pattern difference of the power spectrum according to the division. Then, the difference metric of the power spectrum pattern difference of inter-sample sets is calculated by dynamic time warping distance, and the consistency metric of the power spectrum pattern difference of intra-sample sets is calculated by mean pooling, and the optimal time-frequency sub-band is selected based on these two metrics. Finally, the optimal time-frequency sub-bands are respectively used for the common spatial pattern feature extraction and the linear kernel support vector machine classification. The proposed PSPD method achieves an average classification accuracy of 84.51%, 84.10% and 73.21% in BCI Competition IV Dataset IIa, Dataset IIb and OpenBMI dataset, respectively. Compared with the state-of-the-art methods, the PSPD method achieves the highest classification accuracy and good stability on three datasets. Excellent experimental results show that the PSPD method has great potential in MI-BCI's motion intention decoding.

Index Terms—Brain-computer interface (BCI), electroencephalogram (EEG), motor imagery (MI), time-frequency selection, power spectrum pattern difference.



I. INTRODUCTION

BRAIN-COMPUTER interface (BCI) is a new technology, which constructs the direct information pathway between a human brain and the external environment by decoding the specific activity pattern of the brain [1]. BCI does not rely on any muscle or peripheral nervous system to interact with the outside world, so it has great potential in the field

of medical rehabilitation [2]. Some excellent applications developed with the help of BCI technology, such as exoskeletons [4], [5], orthotics [6] and wheelchairs [7], can help patients with neuromuscular disorders to restore impaired motor functions.

Electroencephalogram (EEG) is widely used to record neural activity in the brain because it is non-invasive, low latency, and safe [8]. During motor imagery, sensorimotor rhythm (SMR) is the specific neuronal activity recorded in the sensorimotor cortex of the brain, which has been proved to be strongly correlated with motor intention or body intention [9]. It is the most typical paradigm in BCI to decode the subject's movement intention by detecting the SMR in the EEG signal. In the motor imagery paradigm, when the subject imagines the movement of the limbs on different sides of the body, the SMR on the opposite side of the moving limb will have a regular power decrease, and the SMR on the same side of the moving limb will have a power increase. This phenomenon mainly occurs in the μ (8-13Hz) and β (14-30Hz) bands, and is also called event-related desynchronization/synchronization (ERD/ERS) [10], [11]. Therefore, by detecting movement-related SMR activities, EEG-based MI-BCI can decode

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different types of neural activities in the brain, and then convert the brain's intention into computer command that can help the subject control external devices.

Existing BCI technology studies have shown that motor patterns can be decoded by sensorimotor rhythms generated by ERD/ERS patterns. Chaisaen *et al.* [12] found that standing and sitting movements during action observation (AO) and motor imagery (MI) have good performance in neural intention decoding, with the potential to be applied to exoskeleton rehabilitation. Jeong *et al.* [13] proposed a subject-dependent section-wise spectral filtering method, which decoded the user's motion intention by detecting the MRCP of the lower limbs and verified the feasibility of autonomous walking on a BMI-based powered exoskeleton. Due to the differences in the subjects' motor imagery ability and the delayed response to task cues, the ERD/ERS patterns activated by motor imagery occurred in a specific time window and within a specific frequency band [14], [15]. In addition, the EEG signal has the characteristics of non-stationary and low signal-to-noise ratio, which makes the signal easily polluted by various noises [16], [17]. Therefore, it is necessary to analyze the expression of sensorimotor rhythms in the temporal and frequency domains to extract efficient features in reliable EEG time-frequency bands and improve the high-precision decoding of EEG signals.

Currently, many motion intention decoding algorithms have been developed to improve the classification performance of EEG [16]–[20], [26]–[28]. Common spatial pattern (CSP) is a spatial filtering feature extraction method, which extracts discriminative features by maximizing the variance of two classes of signals and finding a set of optimal spatial filters [18]. CSP algorithm usually uses a wide frequency band (8–30Hz) and a fixed time window (0.5s–2.5s) to extract features, but the fixed frequency band and time window limit the improvement of classification accuracy [19]. Novi *et al.* [20] proposed an improved method called Sub-band Common Spatial Pattern (SBCSP), which uses filter banks to divide EEG signals into sub-bands for feature extraction and classification, achieving higher classification performance than CSP. Jin *et al.*, aiming at the problem that the CSP algorithm is very sensitive to noise and has low generalization ability, proposed a series of improved algorithms, such as the internal feature selection based on the difference and ratio of average L1-norm for CSP (DRL1-CSP) method [21], the new regularized CSP (RCSP) method [22], the iterative minimum covariance determinant (IMCD) algorithm [23], the time filter and (e)TRCA-based similarity measurement method [24]. The proposed method can obtain more efficient CSP features and improve classification performance. Wang *et al.* [25] found that the MRCP and ERD features of pre-movements had significant complementary discriminative information. Therefore, they propose a discriminative canonical pattern matching (DCPM) method and combine the DCPM method with the CSP method to achieve good pre-action recognition. Zhang *et al.* [26] proposed a temporally constrained sparse group spatial pattern (TSGSP) method to select the best time-frequency sub-band by sparse opti-

mization of CSP features. Das *et al.* [27] proposed a cognitive fuzzy inference system (CFIS) method to find the subject-specific spatial spectrum filter by constructing a five-layered interval type-2 neuro-fuzzy inference system and then obtaining the subject's most suitable frequency band. Xu *et al.* [28] proposed a time-frequency selection method based on particle swarm optimization, which encodes the frequency band and time interval with particles, and then optimizes the CSP feature to obtain the best time-frequency band of the subject.

However, all of the above studies are optimized for the optimal frequency and time segments at the feature level, and do not make full use of the ERD/ERS pattern change information related to motor imagery. The spectral feature information from time segments and frequency bands is usually used as a representative feature of EEG, and it has been proven to be a reliable marker in many EEG classification algorithms [29], [30]. Suk *et al.* [31] proposed a time-frequency graph analysis method based on event-related spectral perturbation (ERSP), which uses the ERSP graphs to measure the intensity of ERD and ERS patterns, and then selects the best frequency band for the subject by setting specific threshold selections. Guerrero-Mendez *et al.* [32] proposed a power-based connectivity (PBC) method, which correlates EEG channels in a specific time segment through Spearman correlation coefficients, and extracts effective power spectrum features for LDA classifier classification. Yang *et al.* [33] proposed a time-frequency selection method based on a criterion called time-frequency discrimination factor (TFDF), which analyzes the variance of EEG signals and selects the time-frequency sub-band when event-related desynchronization occurs. However, the above methods only make use of simple information in the frequency domain, and do not take into account the complex and changeable information between sample sets and the interaction between brain regions.

In this paper, we propose a power spectrum pattern difference-based time-frequency sub-band selection (PSPD) method to further improve the classification accuracy of MI-EEG, and the framework of PSPD algorithm is shown in Fig. 1. Specifically, the raw EEG signal is divided into two data subsets after filtering the noise components, and then the power spectrum information of the two subset signals is extracted, respectively. To better analyze the SMR, we divide the EEG channels and extract the average pattern difference and the maximum pattern difference of the power spectrum according to the division. Then, the difference metric of the power spectrum pattern difference of inter-sample sets is calculated by dynamic time warping distance, and the consistency metric of the power spectrum pattern difference of intra-sample sets is calculated by mean pooling, and the optimal time-frequency sub-band is selected based on these two metrics. Finally, CSP is used to extract the discrimination feature of the optimal time-frequency sub-band, and the linear kernel SVM is used to accurately identify the MI task. The proposed PSPD method has been extensively evaluated on datasets IIa, dataset IIb and OpenBMI dataset, and compared with the state-of-the-art methods. Results show that

our proposed PSPD method has more excellent classification performance. The main innovations and contributions of this work are presented below:

- 1) We innovatively proposed to select subject-specific time-frequency subsets based on the power spectrum pattern differences of inter-and intra-sample sets.
- 2) The subject-specific time-frequency subset selected by the proposed PSPD method can obtain more separable CSP features.
- 3) The PSPD algorithm has been extensively evaluated on the BCI competition IV dataset IIa, dataset IIb and OpenBMI dataset, and the average classification accuracy of 84.51%, 84.10% and 73.21% is obtained, which is the most competitive among existing methods.

The remaining part of this paper is organized as follows: Section II explains the proposed PSPD method. Section III describes the datasets used in the experiments and experimental protocol. Section IV illustrates the experimental results, Section V discusses the results of the paper, future work and limitations. Finally, Section VI presents the conclusions.

II. METHODS

In this section, we first describe the preprocessing method of the raw EEG signal. Then, the proposed PSPD method is described in detail. Finally, the feature extraction method and feature classification method are introduced.

A. Preprocessing

In the preprocessing stage, we use the fifth-order Butterworth notch filter to filter out the power line interference contained in the raw EEG signal, and then use a 4-40Hz fifth-order Butterworth bandpass filter to remove high-frequency noise above 40Hz and low-frequency noise below 4Hz. The ERD/ERS phenomena generated by motor imagery are mainly distributed in the α and β bands of EEG signals, so the frequency range of bandpass filtering needs to include α and β bands. Since the response time of subjects to the cues varies from person to person, the recorded EEG data during the MI tasks contain many redundant segments. Therefore, it is necessary to select an appropriate time window to eliminate time segments irrelevant to the task and retain time segments activated by ERD/ERS.

In view of the advantages of the SBCSP and TSGSP methods, the frequency bands and the time of filtered signals are divided into several sub segments. The raw multi-channel EEG signal in each trial can be defined as $X \in \mathbb{R}^{c \times s}$, where the c is the number of channels and the s is the number of sample points in one trial. For each subject, there are n trials in the MI datasets. The frequency range of ERD/ERS pattern of each channel is divided into f sub-bands by fifth-order zero-phase Butterworth bandpass filters, each band has a bandwidth of 4Hz with a 2Hz overlap. The X is also divided into k time windows, the period of each window is 2s with 0.5s overlap.

B. Power Spectrum Pattern Difference-Based Time-Frequency Sub-Band Selection Method

When the subject imagines the movement of different parts of the body, the EEG recorded by the brain sensorimotor

cortex will occur ERD/ERS phenomenon, that is, the power of the sensorimotor rhythm on the ipsilateral side of the limb will decrease, and the power of the opposite side of the limb will increase. Therefore, we can use this rule to identify MI tasks. In this paper, the short-time Fourier transform (STFT) is employed to obtain the power spectrum of the signal. For each trial, the STFT is applied for each channel. Hence, we can obtain the power spectrum $\mathcal{X}^i \in \mathbb{R}^{c \times (p \times q)}$ of the i -th trial signal as follows:

$$\mathcal{X}^i = \left[STFT(x_1^i), STFT(x_2^i), \dots, STFT(x_c^i) \right], \quad (1)$$

where, the x_j^i is the j -th row vector of the X^i . The $STFT(\cdot)$ is the STFT function. The p and q represent the frequency resolution and the time resolution of the STFT power spectrum respectively. The p is fixed to 1024 and the q is configured according to the sliding time window of STFT. Since the spectrum should be aligned to the bands and time windows in the preprocessing section, the STFT power spectrum have to be resized. Assuming that, the f_l and the f_o are the length and the overlap length of the frequency sub-bands, the k_l and the k_o are the length and the overlap length of the time windows. The bi-cubic interpolation is employed to resize the dimension of $\mathcal{X}^i$ to $\hat{\mathcal{X}}^i \in \mathbb{R}^{c \times (\hat{p} \times \hat{q})}$, where

$$\hat{p} = \left\lceil \frac{p}{f \cdot (f_l - f_o) + f_o} \right\rceil \cdot (f \cdot (f_l - f_o) + f_o), \quad (2)$$

$$\hat{q} = \left\lceil \frac{q}{k \cdot (k_l - k_o) + k_o} \right\rceil \cdot (k \cdot (k_l - k_o) + k_o). \quad (3)$$

where $\lceil \cdot \rceil$ represents the symbol of ceiling. After resizing, the power spectrum can be simply aligned with the time-frequency sub-band in the preprocessing section one by one, that is to say, it is easy to divide the power spectrum into $f \times k$ time-frequency sub-bands. In the subsequent feature extraction process, we will select the effective time-frequency sub-bands to obtain effective features through the power spectrum pattern difference of time-frequency sub-bands.

When subjects performed the left-hand and right-hand MI task, regular ERD/ERS phenomena occurred in the sensorimotor cortex on the left and right sides of the brain. Inspired by this phenomenon, we attempted to partition the sensorimotor cortex and merge adjacent electrode channels in an average way to better analyze the sensorimotor rhythm changes on different sides. When recording the EEG signals of the subjects, the location of the brain electrodes is arranged in strict accordance with the international 10-20 system. As shown in Fig. 1, we divided the brain electrodes along the sagittal line into the middle brain electrode group set M , the left-brain electrode group set L , and the right-brain electrode group set R . The number of elements in set M , L and R are m , l and r respectively. The power spectrum of each channel group $\mathcal{P}^i$ is the average spectrum of channels in the channel group set as follows:

$$\mathcal{P}_L^i = \frac{1}{l} \sum_{j \in L} \hat{\mathcal{X}}_j^i, \quad \mathcal{P}_M^i = \frac{1}{m} \sum_{j \in M} \hat{\mathcal{X}}_j^i, \quad \mathcal{P}_R^i = \frac{1}{r} \sum_{j \in R} \hat{\mathcal{X}}_j^i. \quad (4)$$

where, the $\hat{\mathcal{X}}_j^i \in \mathbb{R}^{\hat{p} \times \hat{q}}$ is the j -th matrix of $\hat{\mathcal{X}}^i$ along channel dimension. After area division, we have 3 power spectrum

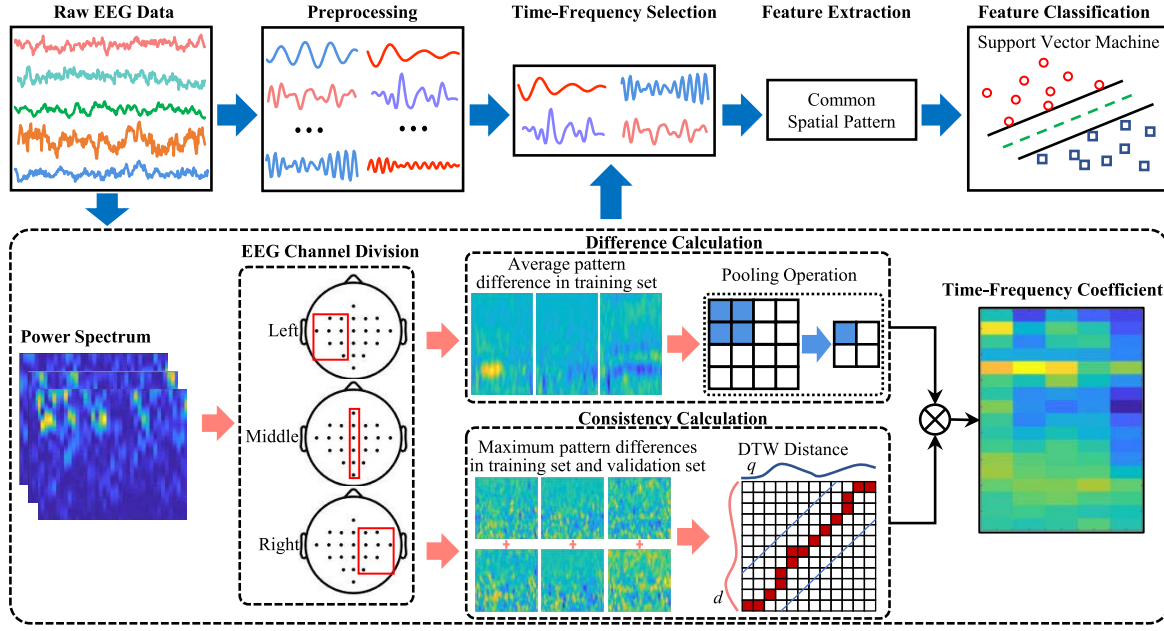


Fig. 1. Framework of the power spectrum pattern difference-based time-frequency sub-band selection method.

$\mathcal{P}_L^i$, $\mathcal{P}_M^i$ and $\mathcal{P}_R^i$ in each trial. The dimension of channels are reduced.

To evaluate the differences of power spectrum in different MI pattern, we divide the dataset into the train set T , the validation set V and the test set. These three sets have no overlap. Only the T set are used to train the CSP projection matrix. We divide the training set T and the validation set V into data sets T_1 , T_2 , V_1 , V_2 that contain only one class according to the class of MI patterns, and are defined as follows:

$$T_1 := \{i \mid (i \in T) \cap (i \in \lambda_1)\}, \quad (5)$$

$$T_2 := \{i \mid (i \in T) \cap (i \in \lambda_2)\}, \quad (6)$$

$$V_1 := \{i \mid (i \in V) \cap (i \in \lambda_1)\}, \quad (7)$$

$$V_2 := \{i \mid (i \in V) \cap (i \in \lambda_2)\}. \quad (8)$$

where λ_1 and λ_2 represent the classes of MI tasks.

For different classes of MI tasks in the dataset, we hope to find the appropriate time-frequency band that maximizes the signal difference between the two classes. According to the ERD/ERS phenomenon, the same changes in EEG signal power in sensorimotor cortex regions occur for the same class of MI tasks. To strengthen the variation of the power spectrum in the same class, we integrate the MI task trials of the same class in the data set by averaging, and then make the difference to obtain the average power spectrum pattern difference. Hence, the pattern differences $\mathbf{D} \in \mathbb{R}^{3 \times (\hat{p} \times \hat{q})}$ of different MI pattern can be calculated as follows:

$$\mathbf{D} = \left| \frac{1}{|\{T_1\}|} \sum_{j \in T_1} \mathcal{P}_{\{L,M,R\}}^j - \frac{1}{|\{T_2\}|} \sum_{h \in T_2} \mathcal{P}_{\{L,M,R\}}^h \right|, \quad (9)$$

in which, the $|\{T_1\}|$ and $|\{T_2\}|$ are the number of elements in the set T_1 and T_2 . The average power spectrum pattern difference of the subject 7 in dataset 2b is shown in Fig. 2.

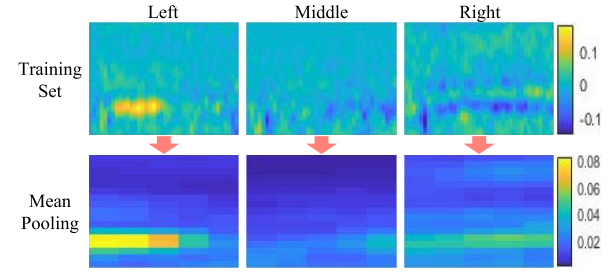


Fig. 2. The result of the pooling operation. The first row is the average power spectrum pattern difference of the left, middle, and right brain electrode groups, and the second row is the result of the pooling operation.

By calculating the power spectrum pattern difference between the classes, the time-frequency position with the largest difference between the classes can be obtained. And by calculating the power spectrum pattern difference in an average way, the positions with large differences in the power spectrum pattern can be more concentrated (the yellow and dark blue areas in the figure). In order to more stably represent the power spectral pattern differences in the time-frequency sub-bands, we use the mean pooling method to average the power spectral pattern differences in each time-frequency sub-band. After mean pooling, the pattern differences of time-frequency sub-bands $\hat{\mathbf{D}} \in \mathbb{R}^{3 \times (\hat{f} \times \hat{k})}$ can be obtained. As shown in the result of the pooling operation in the second row of Fig. 2, the time-frequency sub-bands with obvious differences will be more prominent.

The pattern differences are obviously, however, only the power spectrum pattern differences are not enough to ensure the quality of the time-frequency sub-bands because of the differences may be caused by artifacts or noise. For the training set and validation set, we hope that within the selected time-frequency sub-bands, the inter-class differences of EEG signals should be as consistent as possible. That is, in the selected time-frequency sub-band, the training set and the

verification set have the same inter-class difference. Therefore, we introduce a consistency metric to measure the similarity between two data sets. Assuming that the EEG signals of training set T and validation set V are similar in a time-frequency band, the power spectral pattern difference between the two data sets should be very small. We define the power spectral pattern difference between sets as Δ , then we have the following equation:

$$T + \Delta = V. \quad (10)$$

Our goal is to minimize the difference Δ between sets as much as possible. By taking the peak value of the power spectral pattern in the set and then calculating the pattern difference, the mutation information of the power spectral pattern in the EEG signal can be better preserved. At first we define the inner set maximum function M as follows:

$$M(S) = \begin{bmatrix} \max_{j \in S} \mathcal{P}_{\{L,M,R\}}^j(1, 1), & \dots, & \max_{j \in S} \mathcal{P}_{\{L,M,R\}}^j(1, \hat{q}) \\ \vdots & & \vdots \\ \max_{j \in S} \mathcal{P}_{\{L,M,R\}}^j(\hat{p}, 1), & \dots, & \max_{j \in S} \mathcal{P}_{\{L,M,R\}}^j(\hat{p}, \hat{q}) \end{bmatrix},$$

where, the S is the input set. This function calculate the maximum power spectrum of set S , so the results of this function $M(S) \in \mathbb{R}^{3 \times (\hat{p} \times \hat{q})}$. The maximum pattern differences in train and validation set, $M^T \in \mathbb{R}^{3 \times (\hat{p} \times \hat{q})}$ and $M^V \in \mathbb{R}^{3 \times (\hat{p} \times \hat{q})}$, are defined as:

$$M^T = |M(T_1) - M(T_2)|, \quad (11)$$

$$M^V = |M(V_1) - M(V_2)|. \quad (12)$$

The M^T and M^V represent the maximum differences of set T and V . It can be imagined that if the signal in a time-frequency sub-band is effective, the difference between the patterns in the train set and the validation set should be similar. On the contrary, this segment is more likely to be affected by noise. Hence, we introduce the DTW distance of each time-frequency sub-bands in T and V to evaluate the signal consistency. DTW distance can find the best matching path between data in two sequences by adjusting the corresponding relationship between time points, and can effectively measure the similarity between sequences. Since the M^T and M^V are aligned to the time-frequency window in preprocessing section, the M^T and M^V can be easily divided to $f \times k$ segments. Hence we can obtain the r -th frequency band and the s -th time window of M^T and M^V with flattening operation which can be defined as

$$\hat{M}^T(r, s) \in \mathbb{R}^{3 \times (1 \times \frac{\hat{p} \cdot f_1}{f \cdot (f_1 - f_0) + f_0} \times \frac{\hat{q} \cdot k_1}{k \cdot (k_1 - k_0) + k_0})} \text{ and } \hat{M}^V(r, s) \in \mathbb{R}^{3 \times (1 \times \frac{\hat{p} \cdot f_1}{f \cdot (f_1 - f_0) + f_0} \times \frac{\hat{q} \cdot k_1}{k \cdot (k_1 - k_0) + k_0})}, \quad r \in [1, f], s \in [1, k].$$

The DTW distance function of the maximum differences of set T and V can be defined as:

$$W(r, s) = DTW(\hat{M}_{\{L,M,R\}}^T(r, s), \hat{M}_{\{L,M,R\}}^V(r, s)), \quad (13)$$

so the DTW distance matrix $W \in \mathbb{R}^{3 \times (f \times k)}$ can be defined as:

$$W = \begin{bmatrix} W(1, 1), & \dots, & W(1, k) \\ \vdots & & \vdots \\ W(f, 1), & \dots, & W(f, k) \end{bmatrix}. \quad (14)$$

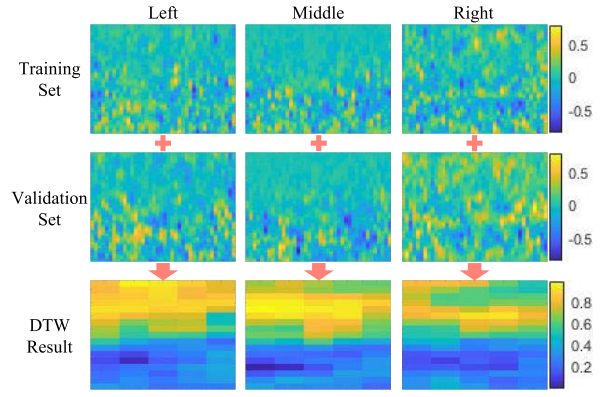


Fig. 3. Pattern consistency results. The first row and the second row are the maximum power spectrum pattern difference of the training set and the validation set, respectively, and the third row is the result of pattern consistency.

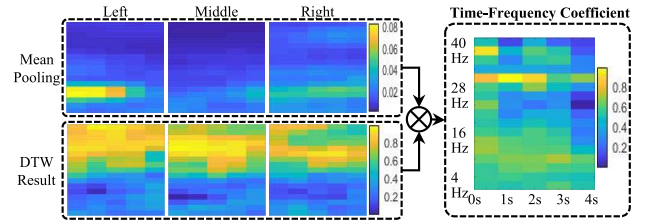


Fig. 4. Schematic diagram of the results of time-frequency coefficients. The time-frequency coefficients are obtained by fusion of mean pooling and DTW results.

For the detailed DTW distance calculation process, please refer to Section A of the Supplementary Material.

Since the amplitude of the EEG signal is very weak, the calculated DTW distance value is very small. Besides, The smaller the distance, the better the consistency, and vice versa. So we take the negative sign of the DTW distance and scale it to the range of 0 to 1 channel by channel and obtain the normalized DTW distance matrix $\hat{W}$. The pattern consistency of the subject 7 in dataset 2b is shown in Fig. 3. As shown in Fig. 3, the maximum power spectrum pattern difference distributions of the training set and the validation set are relatively discrete, which makes it difficult to distinguish the best time-frequency sub-bands. Through the DTW distance calculation, the result obtained can better distinguish the pattern similarity between different time-frequency sub-bands.

After the pattern differences and consistency are obtained, the time-frequency coefficient $C \in \mathbb{R}^{f \times k}$ can be defined as follows:

$$C = \sum_{j \in \{L,M,R\}} \hat{D}_j \odot \hat{W}_j. \quad (15)$$

where $\odot$ represents the Hadamard product. A threshold ε is defined to select the time-frequency sub-bands, all the coefficients which is upper than ε are selected.

The time-frequency coefficient of subject 7 in data set 2b is shown in Fig. 4. The results of mean pooling show that the power spectrum pattern difference is mainly distributed in the low-frequency band, and the results of consistency show that the consistency between sample sets is better in the high-frequency band. The time-frequency coefficient C obtained

by fusing the difference and consistency can retain the time-frequency difference information of the low-frequency band, provide the consistency information of the high-frequency band, and will not overwhelm the original information.

C. Feature Extraction

Common spatial pattern (CSP) is one of the most commonly used feature extraction algorithms in MI EEG signal classification, which is mainly used for spatial filtering processing under two classification tasks. CSP algorithm projects the signal by finding a set of optimal spatial filters, which minimizes variance difference of the same class and maximizes variance difference between different classes, and the definition of the filter is as follows:

$$J(\omega) = \frac{\omega^T \overline{R_1} \omega}{\omega^T \overline{R_2} \omega}, \quad (16)$$

where $\overline{R_1}$ and $\overline{R_2}$ represent the average covariance of all trials in class 1 and class 2, respectively. Eq. 16 can be solved by generalized eigenvalue decomposition. By sorting the eigenvectors, the optimal spatial filter ω is obtained by selecting the eigenvectors corresponding to the four maximum and minimum generalized eigenvalues. For a given motor imagery EEG data e , the CSP feature vector $Z = [z_1, z_2, \dots, z_{2a}]$ is calculated as follows:

$$z_i = \log(\text{var}(\omega_i^T)) \quad (17)$$

where $\text{var}(\cdot)$ represents the variance.

D. Feature Classification

Support vector machine (SVM) is a supervised algorithm, which maps the input data to a high-dimensional space through the kernel trick, and then separates the data through the hyper-plane with the largest margin. In this work, we concatenate the extracted CSP features into a feature vector, and then input it into a linear kernel support vector machine classifier to recognize the motor imagery task. We use the `fitsvm` function in MATLAB2019a to train linear kernel SVM classifier with default kernel parameters.

III. EXPERIMENTAL STUDY

A. EEG Data Description

In order to better evaluate the effectiveness of our method in the classification of MI EEG signals, three public and commonly used data sets will be used to test the classification performance of our method.

1) *BCI Competition IV Data Set IIa*: This dataset recorded 22 channels of EEG signal data during the motor imagery period of 9 subjects at a sampling rate of 250 Hz. During the experiment, each subject was required to perform the MI task for 3s according to the task cues. Each subject contained two sessions. In each session, four MI tasks (left hand, right hand, foot, and tongue) were randomly performed, with a total of 288 trials (72 for each MI task). More details about the dataset could be found in <http://www.bbc.de/competition/iv/>.

2) *BCI Competition IV Data Set IIb*: This dataset contained 9 subjects, all right-handed. EEG signals of C3, C4 and Cz channels were recorded at 250 Hz for left hand and right hand MI tasks. Only the third training session of the dataset is used in this study, i.e., “B0103T”, “B0203T”, ..., “B0903T”. For each subject, a total of 160 MI task experiments with smile feedback were recorded (80 for left-hand MI and 80 right-hand MI). For each trial, there is a 3.5s preparation stage, followed by a 4.5s MI visual cue, informing the subjects that they need to perform a left-handed or right-handed MI task, and finally has a random rest time of 1-2s. More details about the dataset could be found in <http://www.bbc.de/competition/iv/>.

3) *OpenBMI Dataset*: This dataset recorded 62-channel EEG data of 54 subjects (25 females, 29 males) with a sampling frequency of 1000Hz. The dataset contains EEG data of two sessions, each session consists of a training phase and a testing phase, and each phase has 100 MI tasks (50 for the left and right hands). In the experiment, subjects were required to perform left-hand or right-hand MI tasks for 4s according to visual cues. In this paper, we only use data from session 1 in the motor imagery paradigm, and select 20 channels that are the same as in the paper [34], [35] (FC-5/3/1/2/4/6, C-5/3/1/z/2/4/6, and CP-5/3/1/z/2/4/6).

B. Experimental Protocol

To better evaluate the classification performance of our proposed PSPD method, we conducted extensive evaluation among the following algorithms:

- 1) STF: Use a time-frequency band with a frequency band of 4-40 Hz and a time window of 0.5-3.5 s.
- 2) MTF: Use a time-frequency subset composed of multiple frequency bands and multiple time windows.
- 3) PSPD-R: Select the time-frequency sub-band opposite to the PSPD method.
- 4) PSPD: Use the best time-frequency sub-band selected by the PSPD method.

After applying each of the above comparison methods, we use the CSP algorithm to extract the features of the time-frequency sub-bands and set the filter pairs of the CSP algorithm to 1, and then use the linear kernel support vector machine classifier to identify the MI task.

In the preprocessing stage, we divide the frequency bands of the EEG signal into $F = 17$ sub-bands, each of which has a bandwidth of 4 Hz and a step size of 2 Hz, that is, 4 – 8 Hz, 6 – 10 Hz, ..., 36 – 40 Hz. The raw EEG signal is decomposed into F sub-bands through a fifth-order zero-phase Butterworth bandpass filter. For time window division, we divide the period for performing the MI task into time windows with a window length of 2s and a step length of 0.5s. For dataset IIa, it is decomposed into $T = 3$ time windows, that is 0 – 2s, 0.5 – 2.5s, 1 – 3s. For dataset IIb, it is decomposed into $T = 5$ time windows, that is, 0 – 2s, 0.5 – 2.5s, ..., 2 – 4s. The OpenBMI dataset has the same setup as dataset IIb.

In the PSPD method, the input multi-channel EEG signal needs bandpass filtering between 4 and 40Hz, which is used to filter out the signal components unrelated to the MI task events. In dataset IIa, channels 2, 3, 7, 8, 9, 14, 15, 19 were

divided into left-brain electrode group, channels 1, 4, 10, 18, 20, 22 were divided into middle brain electrode group, and channels 5, 6, 11, 12, 13, 17, 18, 21 were divided into right-brain electrode group. There are only three channels in dataset IIb, and these three channels are located in the left, middle, and right regions of the brain. Meanwhile, to better correspond to other datasets, we define the C3, Cz, and C4 channels as the left, middle, and right electrode channels, respectively. The value of the hyperparameter ε is determined by cross-validation of training data.

We used a 10×10 -fold cross-validation method to calculate the classification accuracy of the subjects. The MI task events were randomly divided into ten equal parts, and then each part was successively used as test data, with seven parts for the training set and two parts for the validation set. The classification accuracy of subject i is the average of ten runs, defined as follows:

$$AC_i = \frac{\sum_{j=1}^{10} \sum_{k=1}^{10} C_{jk}}{\sum_{j=1}^{10} \sum_{k=1}^{10} T_{jk}}, \quad (18)$$

where C_{jk} represents the number of trials correctly classified in the j -th calculation of the k -th fold, T_{jk} represents the number of all trials in the j -th calculation of the k -th fold.

IV. RESULTS

A. Subject-Specific Time-Frequency Sub-Band

Due to the huge differences in brain structure, mental state, and response-ability between different individuals, the time window and frequency band of the ERD/ERS pattern varies from person to person. To confirm the importance of optimizing the time window and frequency band of the subjects and the effectiveness of the proposed algorithm, we analyzed the time-frequency sub-bands obtained by the PSPD method on subjects 3, 7, and 9 of dataset IIa, and the visualization result is shown in Fig. 5. We can observe that the PSPD method selects different time window and frequency band combinations for different subjects, which indicates that the power spectrum pattern difference has a big difference between subjects. At the same time, the time-frequency sub-bands with large power spectrum pattern differences of the subjects are not the entire time segment and frequency band during the motor imagery period, which also inspires us to select specific time-frequency sub-bands for subjects to obtain more reliable EEG data segments for capturing the most discriminating features.

B. Comparison of Feature Distribution

To further study the reliability of the optimal time-frequency sub-bands selected by the PSPD method, we reduce the dimensionality of the characteristics of the time-frequency sub-bands, and then analyze them by visualization. The t -distributed stochastic neighbor embedding (t -SNE) method is a nonlinear dimensionality reduction method that maps data in a high-dimensional space into a low-dimensional space while preserving the local properties of the dataset [36]–[38]. We use the t -SNE function in MATLAB 2019a and use the default parameters of the function to reduce the high-dimensional CSP features obtained by PSPD-R, MTF, and PSPD methods

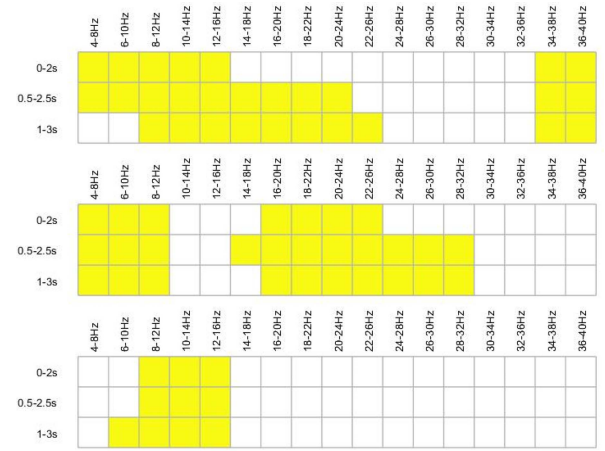


Fig. 5. Visualization results of time-frequency sub-bands selected by the PSPD method. From top to bottom are the subjects 3, 7, and 9 in dataset IIa, respectively, and the yellow squares indicate the selected time-frequency sub-bands.

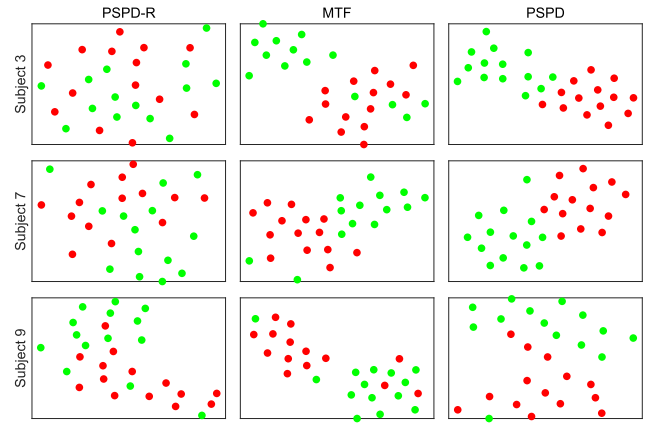


Fig. 6. Visualization results of the feature distribution of PSPD-R, MTF and PSPD methods. From left to right are the PSPD-R, MTF and PSPD methods, and from top to bottom are the feature distribution of subjects 3, 7 and 9 in Dataset IIa, respectively. The red and green dots represent the MI task of left and right hands, respectively.

to a 2-D plane, and the visualization results of the feature distribution are shown in Fig. 6. Note that the PSPD method uses the time-frequency sub-band in the yellow area in Fig. 5, the MTF method uses the time-frequency sub-band in all areas in Fig. 5, and the PSPD-R method uses the time-frequency sub-band in the white area in Fig. 5.

As shown in Fig. 6, we can observe that the CSP feature distribution obtained by using the PSPD method has the best separation, which indicates that the proposed method can find the best time-frequency subset, thereby generating discriminative features. MTF method uses all the time-frequency sub-bands, but it cannot get a good separable feature distribution, so it is necessary to eliminate bad time-frequency sub-bands to improve the classification accuracy. PSPD-R method selects the time-frequency sub-band opposite to the PSPD method, that is, the worst time-frequency sub-band. It can be observed from the figure that the feature distribution obtained by the PSPD-R method is very confusing and difficult to distinguish. The classification performance of CSP depends heavily on the input EEG data, and these poor time-frequency sub-bands will

TABLE I
CLASSIFICATION ACCURACY (%) OBTAINED BY STF, PSPD-R, MTF AND PSPD METHODS IN DATASET IIA, RESPECTIVELY

Subject	Left hand vs. Right hand				Foot vs. Tongue			
	STF	PSPD-R	MTF	PSPD	STF	PSPD-R	MTF	PSPD
A01	76.79	84.64	86.99	91.43	68.04	79.29	78.70	82.86
A02	56.36	54.29	55.83	66.07	85.71	92.14	87.17	90.71
A03	94.50	86.43	94.28	94.64	75.89	62.50	82.93	87.86
A04	67.75	69.64	71.20	75.36	64.61	71.79	75.25	79.64
A05	52.75	74.64	77.85	78.57	62.57	65.00	73.90	75.00
A06	55.14	58.57	59.13	66.79	62.46	75.71	71.27	81.79
A07	77.18	71.79	95.05	97.14	77.68	79.29	86.27	91.07
A08	93.39	92.50	94.05	95.36	88.07	70.00	88.15	89.64
A09	76.25	81.43	84.73	87.86	80.79	78.21	86.80	89.29
Mean±Std	72.23±14.71	74.88±12.06	79.90±14.14	83.69±11.58	73.98±9.34	74.88±8.39	81.16±6.12	85.32±5.38
p-value	<0.01	<0.01	<0.01	-	<0.01	<0.01	<0.01	-

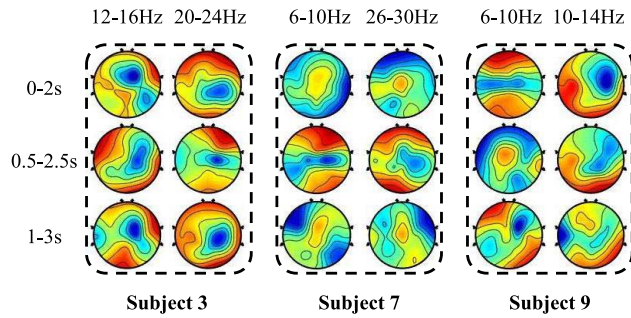


Fig. 7. The channel spectrum and related topographic maps of subjects 3, 7, and 9 in dataset Ila.

make the CSP method project features into a bad feature space, making features more difficult to separate.

C. Channel Spectrum and Related Topographic Maps

To analyze the physiological interpretation of the selected time-frequency sub-bands, we selected two frequency bands in all frequency bands of the three subjects learned by the PSPD method to perform spectral analysis within 0-3s, and the channel spectrum and related topographic maps are shown in Fig. 7. Note that our results are obtained by the functions in the EEGLAB toolbox of MATLAB2019a. The channel spectrum and related topographic maps show the activation of SMR in the brain during the subjects' motor imagery. For subject 3, it can be observed that on the two frequency bands of 12-16Hz and 20-24Hz, SMR is continuously activated within 0-3s of performing motor imagery. For subject 7, it can be observed that SMR activation gradually increased in 0-2s and 0.5-2.5s time windows in the frequency bands of 6-10Hz and 26-30Hz and then decreased in the time window of 1-3s. For subject 9, the SMR continued to increase in the frequency band of 6-10Hz, and then the activation was most obvious in the time window of 1-3s. In the frequency band of 10-14 Hz, SMR is most active in the time window of 0-2s, and then gradually weakened. The above results prove that the proposed PSPD method can effectively capture the subject-specific dynamic ERD/ERS changes in SMR. In the selection of time-frequency sub-band, the PSPD method comprehensively considers the difference metric and the consistency metric. If the consistency result of the time-frequency sub-band is poor, the PSPD method will not select the time-frequency sub-band.

TABLE II
CLASSIFICATION ACCURACY (%) OBTAINED BY STF, PSPD-R, MTF AND PSPD METHODS IN DATASET IIB, RESPECTIVELY

Subject	STF	PSPD-R	MTF	PSPD
B01	70.94	71.25	82.75	85.00
B02	56.63	57.50	50.00	61.25
B03	49.44	50.63	50.63	60.63
B04	96.88	84.38	99.13	100.00
B05	78.94	87.50	90.25	91.25
B06	67.44	79.38	81.19	88.13
B07	79.25	81.88	91.19	90.00
B08	86.75	85.00	91.13	91.88
B09	83.75	81.88	83.31	88.75
Mean	74.44±14.10	75.49±12.34	79.95±16.68	84.10±12.96
p-value	<0.01	<0.01	0.027	-

D. Comparison of Classification Results

Table I and Table II summarize the classification accuracy obtained by the STF, PSPD-R, MTF, and PSPD methods on dataset Ila and dataset Iib, respectively. Note that the selected time-frequency sub-bands are all features extracted by the CSP algorithm, and classified by the linear kernel support vector machine.

The results show that the proposed PSPD method can effectively improve the decoding performance of EEG signals by selecting the best time-frequency sub-band for classification, and the average classification results are 84.51% (Average of 83.69% and 85.32%) and 84.10%, respectively. Compared with STF, PSPD-R and MTF methods, the classification accuracy of the proposed PSPD method on dataset Ila is improved by 11.46%, 8.81%, 3.79% (Left hand vs. Right hand) and 11.34%, 10.44%, 4.16% (Foot vs. Tongue), respectively. And the corresponding classification accuracy on dataset Iib has increased by 9.66%, 8.61% and 4.15%, respectively. To compare the statistical significance of the PSPD method with other competing methods in the classification accuracy, we used the Wilcoxon signed rank test method in the two data sets. As shown in Table I and Table II, the *p*-value of PSPD method and other competing methods in Dataset Ila and Dataset Iib were all less than 0.01, showing statistical significance is very significant (Except for dataset Iib, the Wilcoxon signed rank test result of the PSPD method and MTF is 0.027, and showing statistical significance is significant). In the

TABLE III

COMPARISON OF CLASSIFICATION ACCURACY (%) BETWEEN THE PROPOSED PSPD METHOD AND THE STATE-OF-THE-ART METHODS ON DATASET IIa

Subject	Left hand vs. Right hand				Foot vs. Tongue		
	FBCSP	SW-LCR	TSGSP	PSPD	SW-LCR	TSGSP	PSPD
A01	87.50	86.11	87.00	91.43	70.83	85.00	82.86
A02	57.63	64.58	64.70	66.07	75.00	92.40	90.71
A03	88.88	95.83	93.80	94.64	70.14	89.70	87.86
A04	61.80	64.58	74.30	75.36	63.19	75.80	79.64
A05	77.08	68.06	90.40	78.57	67.36	73.90	75.00
A06	61.11	68.75	63.90	66.79	70.14	75.10	81.79
A07	85.41	81.94	91.40	97.14	79.86	88.30	91.07
A08	92.36	97.22	95.80	95.36	78.47	93.00	89.64
A09	86.11	90.97	81.30	87.86	85.42	82.50	89.29
Mean	77.54	79.78	82.51	83.69	73.38	83.97	85.32
Std(±)	12.90	12.71	11.54	11.58	6.52	7.10	5.38
p-value	<0.01	0.3	0.16	-	<0.01	0.43	-

TABLE IV

COMPARISON OF CLASSIFICATION ACCURACY (%) BETWEEN THE PROPOSED PSPD METHOD AND THE STATE-OF-THE-ART METHODS ON DATASET IIb

Subject	SGRM	FBCSP	NCFS	LRFCSP	T-GRA	TSGSP	PSPD
B01	77.30	76.50	79.25	82.50	78.30	84.00	85.00
B02	59.10	56.82	63.48	60.00	62.70	62.60	61.25
B03	51.50	54.94	56.65	62.78	67.80	56.30	60.63
B04	97.00	99.38	99.28	99.37	97.70	99.40	100
B05	87.40	90.44	88.68	87.50	94.00	94.80	91.25
B06	72.50	79.75	79.96	81.88	87.00	83.80	88.13
B07	86.70	86.50	88.76	90.63	78.00	94.10	90.00
B08	84.70	88.75	92.66	91.25	88.70	93.30	91.88
B09	85.60	81.88	84.95	83.13	89.00	90.10	88.75
Mean	77.98	79.44	81.52	82.12	82.58	84.27	84.10
Std(±)	13.82	14.07	12.94	12.23	11.11	14.16	12.96
p-value	<0.01	<0.01	0.05	0.07	0.57	0.81	-

Foot vs. Tongue task of subject A02, the number of time-frequency sub-bands selected by the PSPD method was small, leading to the relatively large number of time-frequency sub-bands used by the PSPD-R method, which resulted in better results for the PSPD-R. Excellent classification performance shows that the PSPD method has significant advantages over the other competitive methods.

E. Performance Comparison of Proposed Method With the State-of-the-Art Methods

Table III, Table IV and Table V summarize the classification performance comparison between the proposed PSPD method and the state-of-the-art method, respectively. In dataset IIa, compared with the recently reported FBCSP [39], SW-LCR [40] and TSGSP [26] methods, the classification accuracy of the PSPD method has increased by 6.15%, 3.91%, 1.18%, 11.94%, and 1.35%, respectively. In dataset IIb, compared with the recently reported SGRM [41], FBCSP [42], NCFS [43], LRFCSP [44], and T-GRA [45] methods, the classification accuracy of the PSPD method is increased by 6.12%, 4.66%, 2.58%, 1.98%, and 1.52%, respectively.

TABLE V

COMPARISON OF CLASSIFICATION ACCURACY (%) BETWEEN THE PROPOSED PSPD METHOD AND THE STATE-OF-THE-ART METHODS ON OPENBMI DATASET

Method	Accuracy	Std	p-value
CSP_cv	70.10	±16.22	<0.01
CSP	67.26	±18.34	<0.01
CSSP	69.56	±18.96	<0.01
FBCSP	68.83	±19.76	<0.01
BSSFO	67.94	±20.26	<0.01
EEGNet	70.89	±13.01	0.60
DeepConvnet	68.33	± 12.34	0.17
FBCNet	74.70	±13.95	0.72
MIN2Net	61.03	±14.47	-
PSPD	73.21	±17.00	-

Compared with the TSGSP method that requires complex optimization time, the PSPD method does not require a complex optimization process, and the classification accuracy is only reduced by 0.17%, and the standard deviation of the average classification accuracy is lower. In the OpenBMI dataset, the PSPD method achieves higher classification accuracy than the CSP_cv [34], CSP [46], CSSP [47], FBCSP [48], BSSFO [49], EEGNet [50], DeepConvnet [51] and MIN2Net [52] methods. Compared with the FBCNet method [53], the PSPD method only reduced the classification accuracy by 1.49%, but the classification accuracy of 31 out subjects of 54 subjects was higher than the FBCNet method. For detailed classification accuracy of each subject, please refer to Table I of Supplementary material. The results of comparison with the state-of-the-art methods show that the proposed PSPD method has strong competitiveness in the classification algorithms of MI EEG.

V. DISCUSSION

In this work, we propose a power spectrum pattern difference-based time-frequency sub-band selection method (PSPD) to improve the classification performance of MI-EEG. PSPD method analyzes the power spectrum information of the EEG signal to find the activation time window and frequency band of the ERD/ERS pattern during the subject's motor imagery. Specifically, the raw EEG signal is divided into training set and validation set after removing the noise component through filtering, and then the power spectrum information is extracted, respectively. To evaluate the differences of power spectrum in different MI pattern, we divided the channels into left, middle and right brain electrode groups according to the location of EEG channels. Then the power spectrum information based on the EEG channel division will be used to extract the average power spectrum and the maximum power spectrum of different classes. Then, the difference metric of the power spectrum pattern difference of inter-sample sets is calculated by dynamic time warping distance, and the consistency metric of the power spectrum pattern difference of intra-sample sets is calculated by mean pooling, and the optimal time-frequency sub-band is selected based on these two metrics. Finally, the CSP algorithm is used to extract the discriminative features of the best time-frequency sub-bands, and the linear kernel SVM is used to

distinguish the MI tasks. By selecting the subject-specific time window and frequency band, the PSPD method obtained 84.51%, 84.10% and 73.21% classification accuracy on dataset IIa, dataset IIb and OpenBMI dataset, respectively. Compared with competing methods, the Wilcoxon signed rank test results show that the proposed PSPD method has a statistically significant improvement in classification accuracy. Compared with the state-of-the-art methods, the proposed method has better classification performance and stability. In the future, we will apply the PSPD algorithm to online experiments to better evaluate the practicality of the PSPD method in the field of exoskeleton rehabilitation.

Since the specific-subject model means that the BCI system requires a higher cost of computation time, we evaluated the average required computation time of the PSPD method on a desktop computer with a CPU of i7-6700, 16GB RAM, and an environment of MATLAB2019a. The results show that on dataset IIa and dataset IIb, the average required computation time of a single trial of the PSPD method is 83ms and 18ms, respectively, and the average training time cost of a single subject is 113.28s and 16.79s, respectively. A large amount of computation time cost of the proposed algorithm is spent on the 10-fold cross-validation, so the actual average training time cost can be about one-tenth. The test set does not require cross-validation, the calculation time cost of our algorithm on the test set is only a millisecond level, which perfectly meets the real-time requirements of BCI applications. Therefore, the proposed algorithm is expected to be extended to practical applications.

In the process of merging brain electrode channels, we just merge by averaging. However, due to the differences in position and signal strength of brain electrode channels, assigning appropriate weights to each electrode channel by weighting may be a direction to improve the classification performance of the model.

When training the linear kernel SVM, we just used the default hyperparameters. One area that can improve classification performance is to use parametric search methods such as particle swarm optimization, whale swarms, etc. to find the best hyperparameters, but this adds computational time cost to the model. Another consideration is to use better classifiers such as neural network classifiers, random forests, gradient boosting classifiers, etc.

When different training sets and validation sets are divided, the consistency will change slightly. To ensure the stability of the results as much as possible, in our experiments, we adopted the 10×10 -fold cross-validation method. As shown in Table III, the classification accuracy of the PSPD method is higher than that of the FBCSP, SW-LCR, and TSGSP methods, and has a lower standard deviation. In addition, the PSPD method considers the results of consistency and difference comprehensively, and it can be found from the results of the experiments that the randomness has little influence on the selection of time-frequency sub-bands.

In our work, the proposed PSPD method is mainly aimed at the subject-specific classification problem. Currently, subject-independent classification abilities are receiving more and more attention. In particular, the MIN2Net method has shown

good classification performance in subject-independent model classification, which can be further extended for research. In the future, we will try to analyze the classification performance of the PSPD model on subject-independent classification problems to explore the classification performance of the model without calibration.

The STFT method is one of the typical methods in time-frequency analysis [54]. Therefore, STFT is used to analyze EEG signals in our work. In time-frequency analysis, continuous wavelet transform and Stockwell transform are also very effective. Continuous wavelet transform can obtain more accurate local time-frequency characteristics by multi-scale analysis of signal through stretching and translation operation [55]. The Stockwell transform uses a variable-width window function whose window width varies inversely proportional to frequency, resulting in high temporal resolution at high frequencies and high-frequency resolution at low frequencies [56]. Therefore, we can try to replace the STFT method with wavelet transform or Stockwell transform to achieve better classification performance.

Multi-channel EEG signals provide richer spectrum information. The excellent classification accuracy and low computational time cost on dataset IIa and dataset IIb indicate that the proposed PSPD algorithm not only has good classification performance on low-dimensional channels, but also has excellent classification performance on high-dimensional channels. However, the increase in the number of channels also introduces information and noise irrelevant to the task. In future work, time-frequency selection algorithms can be combined with channel selection algorithms to reduce redundant channels and save computational costs to obtain better classification performance.

The limitation of this work is that the performance of the proposed algorithm is very dependent on the choice of threshold ε , which is used to control the selected time-frequency sub-bands. The cross-validation method can obtain better parameter values and produce promising classification accuracy, but it takes a lot of computational time and cost. In future work, we can consider using intelligent parameter optimization methods to obtain the optimal threshold to achieve higher EEG classification performance.

VI. CONCLUSION

In this study, we propose a power spectrum pattern difference-based time-frequency sub-band selection method, which can further improve the classification accuracy of MI EEG. To better analyze the sensorimotor rhythm, we divide the EEG channels and extract the average and maximum pattern difference of the power spectrum according to the partition. Then, the difference metric of the power spectrum pattern difference of inter-sample sets is calculated by dynamic time warping distance, and the consistency metric of the power spectrum pattern difference of intra-sample sets is calculated by mean pooling, and the optimal time-frequency sub-band is selected based on these two metrics. Finally, the CSP algorithm is used to extract discriminative features, and the linear kernel SVM classifier is used to accurately identify the MI task. We conducted an extensive evaluation on dataset IIa,

dataset IIB and OpenBMI dataset to verify the effectiveness of the PSPD method compared with other competing methods. Results show that the classification accuracy of the PSPD method on the three data sets reached 84.51%, 84.10% and 73.21%, respectively. Excellent classification performance verifies that the proposed algorithm can decode the motion intention in EEG efficiently, which is of great significance for developing high-performance BCI.

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